AERO5560 Group 8
Aircraft Modelling and Pitch-Rate Feedback
Progress Report: Data, Trim, Linearisation and Inner-Loop Design

Group 8

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Abstract

Group 8 aircraft data, trim verification, numerical A/B matrices and $G_{q\delta_e}$ support three pitch-rate compensator designs. The selected loop achieves 4.237 rad/s bandwidth, 1.607 s settling, 0.116% error and 8.897% overshoot. At 20 rad/s, open-loop and closed-loop noise magnitudes are -41.390 and -41.409 dB. Nominal targets are met; off-nominal and nonlinear tests define the checked range.

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1 Scope and Modelling Conventions

Longitudinal Task 1 is addressed through aircraft modelling, $G_{q\delta_e}$, a labelled feedback loop and three compensator designs. The vertical-speed loop, gust analysis, auto-throttle and full integration follow.

The full state vector contains body-axis velocities and angular rates, Euler attitude angles, and earth-frame positions:

$$\boldsymbol{X} = [u, v, w, p, q, r, \phi, \theta, \psi, X_e, Y_e, Z_e]^{\mathsf{T}}. \tag{1}$$

The control-input vector is

$$\boldsymbol{U} = [\delta_T, \delta_e, \delta_a, \delta_r, \delta_f]^{\mathsf{T}}, \tag{2}$$

representing throttle, elevator, aileron, rudder and flap, respectively. Internal calculations use SI units: m/s for velocity, m for position, rad for angles, rad/s for angular rates, and a 0–1 throttle fraction. Altitude is $h = -Z_e$, and vertical speed is $V_s = -\dot{Z}_e$, positive upwards.

Linear-model variables are increments about the nominal flight trajectory:

$$\boldsymbol{X} = \boldsymbol{X}_{\text{ref}}(t) + \Delta\boldsymbol{X}, \qquad \boldsymbol{U} = \boldsymbol{U}_0 + \Delta\boldsymbol{U}. \tag{3}$$

The reference trajectory advances in X_e; control variables are increments about $\boldsymbol{U}_0$.

2 Group 8 Aircraft Data

The Group 8 function `FlightDataAircraft8.p` was executed in MATLAB R2025b Update 7. Initial conditions were loaded from `ICsAircraft8_200knts_300ft.mat`.

2.1 Mass, Geometry and Propulsion

Table 1: Principal aircraft parameters extracted from the supplied data

Parameter	Description	Value
m	Mass	750 kg
g	Gravitational acceleration	9.81 m/s^2
S	Reference wing area	11.9 m^2
b	Wingspan	7 m
c	Reference chord	0.85 m
I_{xx}	Roll moment of inertia	605.29 kg m^2
I_{yy}	Pitch moment of inertia	1766.52 kg m^2
I_{zz}	Yaw moment of inertia	2168.83 kg m^2
I_{xz}	Product-of-inertia parameter	84.71 kg m^2
$P_{\max}$	Maximum power parameter	253.504 kW
η	Propeller efficiency parameter	0.85

The corresponding aspect ratio is $b^2/S = 4.11765$, and wing loading is $mg/S = 618.277\,\text{N/m}^2$.

The legacy fields are mapped to the names required by the current nonlinear simulation framework, as instructed in the data package:

```
FlightData = FlightDataAircraft8();
FlightData.Geometric = FlightData.Geo;
FlightData.Inertial = FlightData.I;
FlightData.Propeller = FlightData.Prop;
```

2.2 Initial State and Controls

$$\begin{aligned} \boldsymbol{X}_0 = [&102.8869153573,\ 0,\ 0.4724322704,\ 0,\ 0,\ 0,\\ &0,\ 0.004591730139,\ 0,\ 0,\ 0,\ -91.44]^{\mathsf{T}}, \end{aligned} \tag{4}$$

$$\boldsymbol{U}_0 = [0.7703166691,\ 0.01337052176,\ 0,\ 0,\ 0]^{\mathsf{T}}. \tag{5}$$

The flight condition is listed in Table 2.

Table 2: Initial flight condition and controls

Quantity	Value
$V_0 = \sqrt{u_0^2 + v_0^2 + w_0^2}$	102.888 m/s, approximately 199.99827 knots
$h_0 = -Z_{e0}$	91.44 m, 300 ft
$\alpha_0 = \text{atan2}(w_0, u_0)$	0.263086758°
θ_0	0.263086758°
β_0, ϕ_0, ψ_0	0
p_0, q_0, r_0	0
Throttle	77.0316669%
Elevator	+0.766074467°
Aileron, rudder and flap	0°

Earth-frame vertical speed is

$$V_s = u\sin\theta - v\sin\phi\cos\theta - w\cos\phi\cos\theta. \tag{6}$$

Substituting the Group 8 initial condition gives $V_{s0} = 0$, consistent with level flight and $\theta_0 = \alpha_0$.

2.3 Control Limits

The supplied data specify a throttle range of $[0, 1]$ and elevator, aileron and rudder limits of $\pm 25°$. The assignment explicitly limits the aileron to $\pm 20°$; the stricter assignment limit must be used for lateral performance assessment. No flap-deflection or control-rate limits are provided in these data.

The pitch-rate inner loop must respect the absolute elevator limit:

$$-25° \le \delta_{e0} + \Delta\delta_e(t) \le 25°. \tag{7}$$

Since $\delta_{e0} = 0.766074467°$, the available incremental elevator range is approximately

$$[-25.766074467°,\ 24.233925533°].$$

The remaining positive throttle increment is $1 - 0.7703166691 = 0.2296833309$.

3 Nonlinear Trim Verification

3.1 Method

With zero gust and the original $\boldsymbol{X}_0, \boldsymbol{U}_0$, the supplied `aero5560_aero.p` and `aero5560_motion.p` functions are evaluated. Because the aerodynamic model depends on state derivatives, the following consistency iteration is required:

$$(\boldsymbol{C}_F^{(k)}, \boldsymbol{C}_M^{(k)}) = \mathcal{A}(\boldsymbol{X}_0, \boldsymbol{0}, \dot{\boldsymbol{X}}^{(k)}, \boldsymbol{U}_0, F), \tag{8}$$

$$\dot{\boldsymbol{X}}^{(k+1)} = \mathcal{M}(\boldsymbol{X}_0, \boldsymbol{C}_F^{(k)}, \boldsymbol{C}_M^{(k)}, F). \tag{9}$$

The iteration terminates when the infinity norm of the update is below 10^{-11}, with a maximum iteration count enforced. At the initial condition, convergence is obtained in two iterations, with a final update norm of zero.

3.2 Residuals and Unforced Simulation

Table 3: Initial state-derivative residuals

Quantity	Result
$\dot{u}$	-2.4980×10^{-16} m/s^2
$\dot{v}, \dot{w}$	0 m/s^2
$\dot{p}, \dot{q}, \dot{r}$	0 rad/s^2
$\dot{\phi}, \dot{\theta}, \dot{\psi}$	0 rad/s
$\dot{X}_e$	102.888 m/s, expected forward motion
$\dot{Y}_e, \dot{Z}_e$	0 m/s

The diagnostic tolerances chosen for this check are 10^{-4} m/s^2 for translational acceleration, 10^{-6} in the corresponding SI units for angular acceleration and attitude rates, and 10^{-4} m/s for lateral and vertical earth-frame velocities.

With $\boldsymbol{U}_0$ fixed and no gust or feedback, the full nonlinear model is integrated for 60 s using `ode45`. The first run uses relative tolerance 10^{-9}, absolute tolerance 10^{-11} and maximum step 0.05 s. A tighter check uses 10^{-10}, 10^{-12} and 0.025 s. Outputs are sampled every 0.1 s, yielding 601 points.

Both runs retain the initial speed, altitude, vertical speed, angle of attack, pitch angle and three angular rates. Differences in these outputs are zero in the saved double-precision results. The final forward position is $X_e(60) \approx 6173.28$ m, consistent with $V_0 t$. The supplied trim point can therefore be retained for linearisation. Figure 1 shows the computed histories.

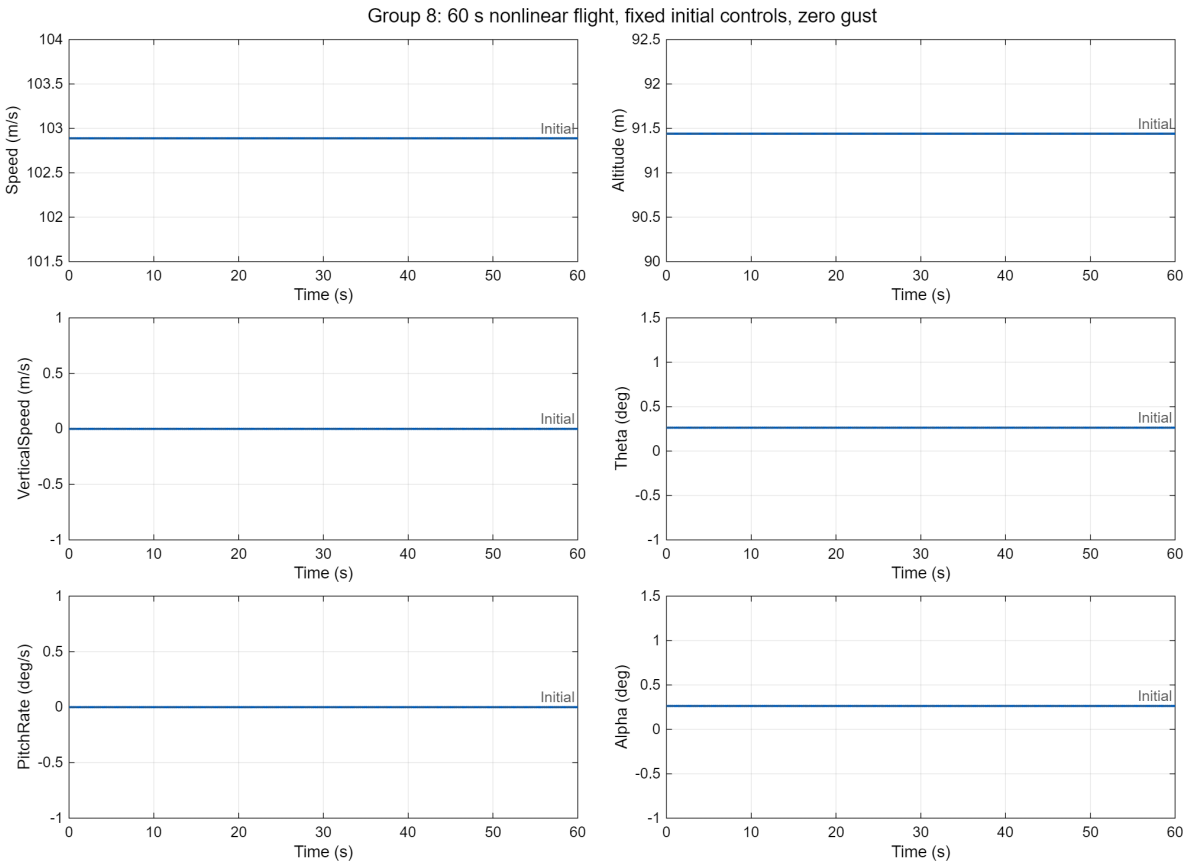


Figure 1: A 60 s nonlinear simulation of Group 8 with fixed initial controls and zero gust.

Stability is assessed below using the linearised poles and perturbed responses.

4 Numerical Linearisation and State-Space Models

4.1 Central Differences and Numerical Checks

Near the nominal trim trajectory, the perturbation model is

$$\Delta \dot{\boldsymbol{X}} = A\Delta \boldsymbol{X} + B\Delta \boldsymbol{U}. \tag{10}$$

The Jacobian columns are calculated using central differences:

$$A_{:,i} \approx \frac{f(\boldsymbol{X}_0 + h_i \boldsymbol{e}_i, \boldsymbol{U}_0) - f(\boldsymbol{X}_0 - h_i \boldsymbol{e}_i, \boldsymbol{U}_0)}{2h_i}, \tag{11}$$

$$B_{:,j} \approx \frac{f(\boldsymbol{X}_0, \boldsymbol{U}_0 + k_j \boldsymbol{e}_j) - f(\boldsymbol{X}_0, \boldsymbol{U}_0 - k_j \boldsymbol{e}_j)}{2k_j}. \tag{12}$$

Every evaluation of f reconverges the implicit state derivatives. State increments are scaled according to velocity, angle/angular-rate and position magnitudes; control scales are unity. Relative steps of $10^{-4}, 3\times10^{-5}, 10^{-5}, 3\times10^{-6}, 10^{-6}$ are compared, and the middle value, 10^{-5}, is selected.

Table 4: Relative Frobenius-norm differences from the selected Jacobians

Relative step	$\|A_h - A\|_F / \|A\|_F$	$\|B_h - B\|_F / \|B\|_F$
10^{-4}	1.1803×10^{-9}	5.1592×10^{-12}
3×10^{-5}	9.5524×10^{-11}	5.5809×10^{-13}
10^{-5}	0	0
3×10^{-6}	1.1302×10^{-11}	1.6652×10^{-11}
10^{-6}	2.3360×10^{-10}	9.7214×10^{-11}

An independent mixed-perturbation Taylor check gives residuals of approximately 5.2498×10^{-5}, 1.3124×10^{-5} and 3.2808×10^{-6} as the perturbation amplitude is successively halved. The error ratios, 4.0002 and 4.0001, agree with the quadratic remainder expected from first-order linearisation.

4.2 Four-State Longitudinal Model

Under the fixed-altitude approximation, state indices $[1, 3, 5, 8]$ and input indices $[1, 2]$ give

$$\boldsymbol{x}_\ell = [\Delta u, \Delta w, \Delta q, \Delta\theta]^\mathsf{T}, \qquad \boldsymbol{u}_\ell = [\Delta\delta_T, \Delta\delta_e]^\mathsf{T}. \tag{13}$$

$$A_\ell = \begin{bmatrix} -0.06144334 & 0.05552868 & -0.47243227 & -9.80989658 \\ -0.16960655 & -4.78731236 & 99.44823052 & -0.04504471 \\ 0.0002475794 & -0.11258862 & -3.67476184 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \tag{14}$$

$$B_\ell = \begin{bmatrix} 2.76318036 & 0 \\ -5.9699623 \times 10^{-5} & -62.90658081 \\ -1.7950804 \times 10^{-5} & -88.19693912 \\ 0 & 0 \end{bmatrix}. \tag{15}$$

The second state is body-axis velocity w.

The four-state longitudinal eigenvalues are

$$\lambda_{\mathrm{SP}} = -4.2313617 \pm 3.3004001\mathrm{i}, \qquad \lambda_{\mathrm{ph}} = -0.03039711 \pm 0.07729208\mathrm{i}. \tag{16}$$

These represent the short-period and phugoid modes, respectively, and lie in the left half-plane.

4.3 Altitude and Lateral-Directional Models

The Z_e column of the full A matrix contains nonzero terms. A five-state longitudinal model retaining altitude uses $[u, w, q, \theta, Z_e]$ and has also been saved. The four-state approximation omits altitude feedback. This report extracts $G_{q\delta_e}$ from Eqs. (14)–(15), consistently using the four-state model.

The lateral-directional states are $[v, p, r, \phi, \psi]$, with aileron and rudder inputs. Their poles are approximately -13.81679, $-0.719193 \pm 2.625197\mathrm{i}$, -0.0119723, and a zero heading-integrator pole. Position and heading include neutral kinematic modes. Full and additional subsystem matrices are provided in the appendices.

4.4 Small-Step Linear and Nonlinear Comparisons

The full 12-state A, B model is compared with the nonlinear model for a $+0.05^\circ$ elevator increment and a $+0.001$ throttle increment over 10 s. The linear response is computed using matrix exponentials and the nonlinear response using `ode45`. Outputs are airspeed, q, and the vertical speed defined in Eq. (6).

Table 5: Maximum absolute output errors over the 10 s comparisons

Input step	Speed (m/s)	q (°/s)	V_s (m/s)
Elevator increment	0.016588	0.023974	0.641690
Throttle increment	2.0867×10^{-6}	1.3892×10^{-7}	2.5237×10^{-7}

For the elevator test, the maximum V_s error during the first 2 s is approximately 0.004968 m/s, or 0.2253% of the nonlinear peak over that interval. By 10 s, however, the vertical-speed peak is approximately 11.93 m/s and the speed change is approximately 4.69 m/s. The aircraft has departed appreciably from the initial point, and the relative V_s error grows to approximately 5.38%. The growing error reflects departure from the trim point.

5 Elevator-to-Pitch-Rate Transfer Function

5.1 Output Selection and Derivation

The second longitudinal input column corresponds to the elevator. Set

$$\begin{aligned} B_q &= B_\ell(:,2) = \begin{bmatrix} 0 & -62.90658081 & -88.19693912 & 0 \end{bmatrix}^{\mathsf{T}}, \\ C_q &= \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}, \qquad D_q = 0. \end{aligned} \tag{17}$$

Taking the Laplace transform with zero initial perturbations gives

$$s\Delta \boldsymbol{x}_\ell = A_\ell \Delta \boldsymbol{x}_\ell + B_q \Delta \delta_e, \tag{18}$$

$$\Delta q = C_q \Delta \boldsymbol{x}_\ell, \tag{19}$$

and hence

$$\boxed{G_{q\delta_e}(s) = \frac{\Delta q(s)}{\Delta \delta_e(s)} = C_q (sI - A_\ell)^{-1} B_q.} \tag{20}$$

Here C_q denotes the state-space output matrix. The complementary sensitivity below is denoted $\mathcal{C}_q$ to avoid ambiguity.

5.2 ZPK and Polynomial Forms

The zero-pole-gain representation, with complex-conjugate pole factors combined into real quadratics, is

$$\boxed{G_{q\delta_e}(s) = \frac{-88.19693912\, s(s + 4.704978073)(s + 0.06347365466)}{(s^2 + 8.462723331 s + 28.79706207)(s^2 + 0.06079421306 s + 0.006898050455)}.} \tag{21}$$

The corresponding polynomial form is

$$G_{q\delta_e}(s) = \frac{-88.19693912 s^3 - 420.5628467 s^2 - 26.33932382 s}{s^4 + 8.523517544 s^3 + 29.31844472 s^2 + 1.809071019 s + 0.1986435871}. \tag{22}$$

The zeros are

$$z = 0, \quad -4.704978073, \quad -0.06347365466. \tag{23}$$

The poles are the longitudinal four-state eigenvalues given above, and the ZPK gain is -88.19693912.

5.3 Numerical Verification and Interpretation

Equation (21) is compared with the state-space frequency response at 501 points from 10^{-4} to 10^{4} rad/s. The maximum relative error is approximately 1.78×10^{-13}. An independently formed Cayley–Hamilton/Markov polynomial gives a maximum relative frequency-response error of approximately 7.20×10^{-9}.

One raw numerical zero is approximately $+2.40 \times 10^{-16}$. The model satisfies $q = \dot{\theta}$, the fourth entry of B_q is zero, and A_ℓ is nonsingular, so an exact zero at the origin is expected. After verifying this structure and the DC gain, only this floating-point residual is set to zero. The raw object is retained; no general pole-zero cancellation is applied.

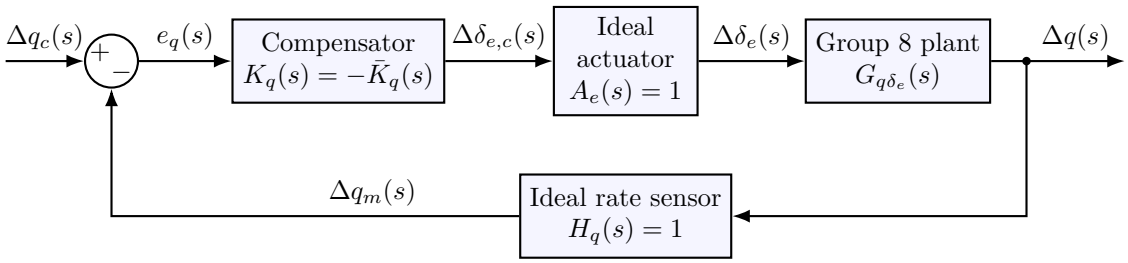
Since $G_{q\delta_e}(0) = 0$, pitch rate returns to zero following a constant elevator input in this stable open-loop model. The leading high-frequency term is approximately $-88.19694/s$, indicating a negative direct pitch-acceleration response to a positive elevator increment. This sign must be handled consistently in the feedback design.

6 Pitch-Rate Feedback-Loop Construction

6.1 Objective and Assumptions

The inner-loop objective is to make the actual pitch-rate increment Δq follow the command Δq_c using the elevator. Unity measurement, $H_q(s) = 1$, and an ideal actuator, $A_e(s) = 1$, are assumed for the linear perturbation design. Physical deflection limits are checked using Eq. (7) and enforced in the nonlinear implementation.

The feedback structure uses the signed compensator $K_q(s)$.



Rate signals: rad/s; elevator signals: rad; all signals are trim increments.

Figure 2: Unity negative-feedback pitch-rate loop using the elevator. The controller includes the required sign; the selected parameters are given below.

Table 6: Signals and blocks in the pitch-rate feedback architecture

Symbol	Definition	Units
Δq_c	Commanded pitch-rate increment	rad/s
Δq_m	Measured feedback, equal to Δq for ideal measurement	rad/s
e_q	Command minus measured pitch rate	rad/s
$K_q(s)$	Signed compensator mapping pitch-rate error to elevator command	s
$\Delta\delta_{e,c}$	Commanded elevator-deflection increment	rad
$A_e(s) = 1$	Ideal actuator: actual increment equals command	Dimensionless
$\Delta\delta_e$	Actual elevator increment applied to the aircraft model	rad
$G_{q\delta_e}(s)$	Open-loop aircraft dynamics from elevator increment to q	(rad/s)/rad
Δq	Actual pitch-rate increment	rad/s
$H_q(s) = 1$	Unity pitch-rate measurement feedback	Dimensionless

6.2 Feedback Sign and Operation

The summing junction implements

$$e_q = \Delta q_c - \Delta q_m. \tag{24}$$

Positive rate error requires negative elevator action because the plant gain is negative. The sign is included in $K_q = -\bar{K}_q$; stability is checked using poles and margins.

The signal relationships are

$$\Delta\delta_{e,c} = K_q e_q, \tag{25}$$
$$\Delta\delta_e = A_e \Delta\delta_{e,c}, \tag{26}$$
$$\Delta q = G_{q\delta_e} \Delta\delta_e, \tag{27}$$
$$\Delta q_m = H_q \Delta q. \tag{28}$$

For example, a 4°/s command corresponds to $\Delta q_c = 4\pi/180 \approx 0.06981317$ rad/s. The comparator, controller and aircraft model all use radians internally to avoid an unintended $180/\pi$ scaling factor.

6.3 Closed-Loop Derivation and SCK Expressions

Define the signed loop transfer function as

$$L_q = K_q A_e G_{q\delta_e} H_q. \tag{29}$$

Combining Eq. (24) with the forward-path relationships gives

$$\Delta q = G_{q\delta_e} A_e K_q (\Delta q_c - H_q \Delta q), \tag{30}$$

Thus the general closed-loop relationships are

$$\frac{\Delta q}{\Delta q_c} = \frac{G_{q\delta_e} A_e K_q}{1 + L_q}, \qquad \frac{\Delta\delta_e}{\Delta q_c} = \frac{A_e K_q}{1 + L_q}. \tag{31}$$

For the assumptions $A_e = H_q = 1$, define sensitivity and complementary sensitivity by

$$S_q = \frac{1}{1 + G_{q\delta_e} K_q}, \qquad \mathcal{C}_q = \frac{G_{q\delta_e} K_q}{1 + G_{q\delta_e} K_q}, \qquad S_q + \mathcal{C}_q = 1. \tag{32}$$

The resulting SCK relationships are

$$\boxed{\frac{\Delta q}{\Delta q_c} = \mathcal{C}_q, \quad \frac{e_q}{\Delta q_c} = S_q, \quad \frac{\Delta \delta_e}{\Delta q_c} = K_q S_q.} \tag{33}$$

The complementary sensitivity $\mathcal{C}_q$ describes command tracking, S_q describes tracking error and responses at appropriate disturbance locations, and $K_q S_q$ gives elevator demand. No noise or gust input is silently introduced at an unspecified point. Subsequent gust transfer functions must be derived for their actual injection locations.

Let N_G, D_G be the numerator and denominator in Eq. (21), and write $K_q = N_K/D_K$. The closed-loop characteristic equation is

$$D_G D_K + N_G N_K = 0. \tag{34}$$

The numerator N_K includes the controller sign. Numerical ZPK forms are saved in the inner-loop folder as `zpk.txt`.

6.4 Trim Bias and MATLAB Construction

The linear loop produces the incremental elevator signal $\Delta\delta_e$. For the nonlinear aircraft, the absolute input is

$$\delta_e(t) = 0.01337052176 + \Delta\delta_e(t) \quad \text{(rad)}, \tag{35}$$

and is limited in the nonlinear implementation.

For the selected signed K_q, the same feedback structure can be constructed in MATLAB:

```
load('Gqde.mat', 'Gqde');
Lq = Gqde * Kq;
Sq = feedback(1, Lq); % error / q_command
Cq_closed = feedback(Lq,1); % q / q_command
Uq = feedback(Kq,Gqde); % elevator_increment / q_command
```

`loop.m` constructs these feedback models for a supplied signed K_q.

7 Pitch-Rate Inner-Loop Design

At 200 knots and 300 ft, the selected bandwidth is 4.237 rad/s and open-loop crossover is 2.017 rad/s. The short-period natural frequency is 5.366 rad/s. Three compensator iterations are:

$$K_{q,1}(s) = -0.1961/s, \tag{36}$$

$$K_{q,2}(s) = \frac{-5.835(s+0.5)}{s(s+0.1)(s+30)}, \tag{37}$$

$$K_{q,3}(s) = \frac{-21.5(s+0.136)(s+0.16)}{s(s+0.0015)(s+0.31)(s+10.7)(s+14.5)}. \tag{38}$$

SISOtool uses $P_q = \text{minreal}(-G/s)$ and $F_q = -sK_q$. A single integrator cancels the plant origin zero in the minimal q channel. The type-0 loop requires $L(0) > 99$ for error below 1%. All controller poles and zeros are real, and the loop is strictly proper.

Table 7: Three compensator iterations

Metric	1	2	3
t_s (s)	10.919	4.546	1.607
$t_{s,command}$ (s)	∞	3.938	1.602
SSE (%)	3.703	0.770	0.116
Overshoot (%)	3.516	16.787	8.897
ω_{BW} (rad/s)	5.211	5.645	4.237
$\lvert L(j20)\rvert$ (dB)	-27.220	-28.885	-41.390
$\lvert T(j20)\rvert$ (dB)	-26.841	-28.595	-41.409

The 4 deg/s step gives absolute elevator 0.4194–0.7661 deg over 0–10 s, phase margin 60.301 deg and gain margin 10.806 dB. The 20 rad/s noise magnitudes satisfy the −40 dB criterion in Lecture 9-1, slides 9 and 17. The checked 20–10^5 rad/s band also passes.

Separate retrimmed 180, 200 and 210 knot points pass the main time-domain and noise targets. An assumed 50 ms actuator gives 15.88% overshoot and passes the main nominal targets. At 100 ms, overshoot is 22.31% and settling is 3.757 s, exceeding both limits. Of 27 parameter cases, 21 meet all time-domain thresholds.

The unreduced system retains a neutral mode because $\dot{\theta} = q$. A 2 s nonlinear pulse reaches 1.759 g and minimum speed 93.874 m/s with fixed throttle. Attitude management and auto-throttle remain necessary.

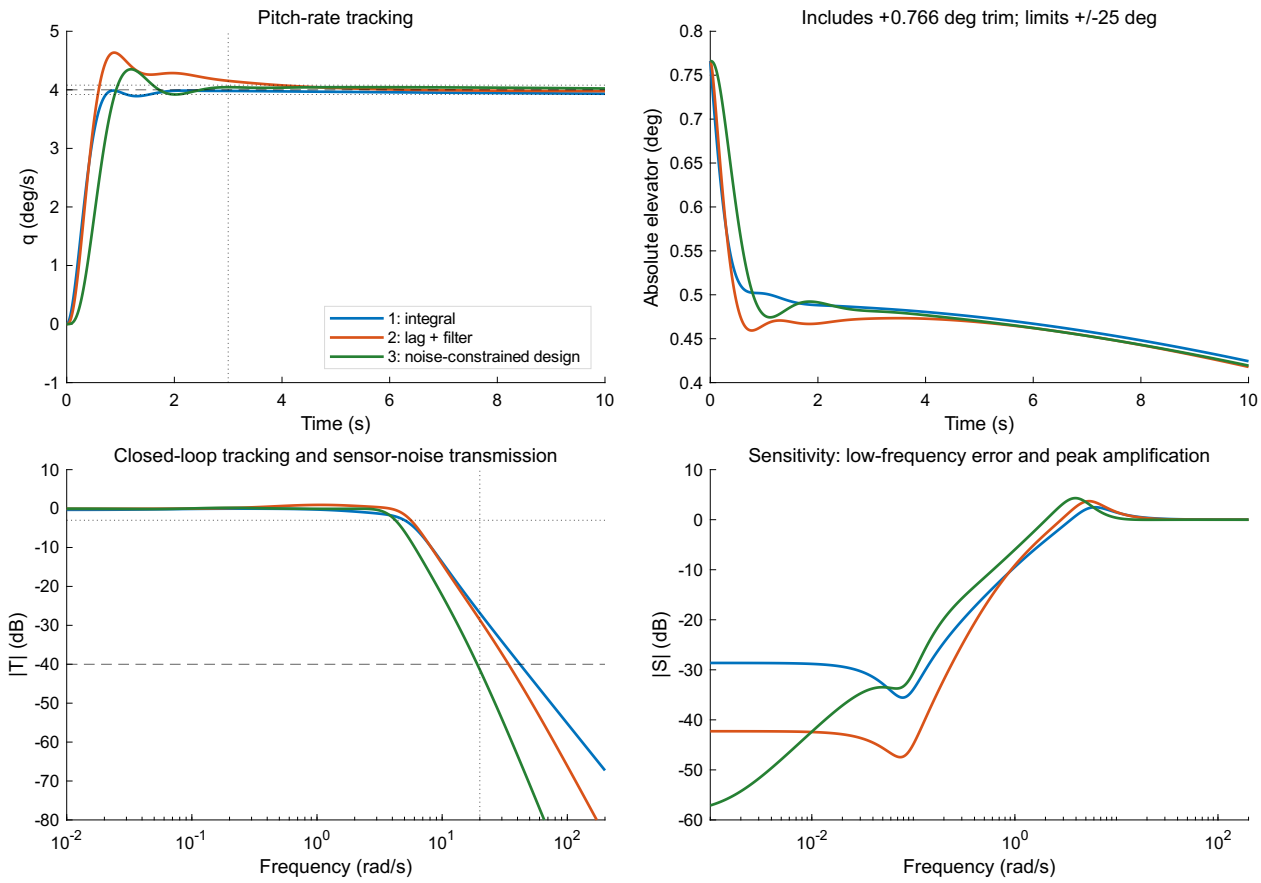


Figure 3: Three iterations: q, elevator, complementary sensitivity and sensitivity.

Full plots and validation are in the inner-loop report. Run `inner_loop/run_inner.m`; inspect using `open_sisotool(3)`.

8 Files, Reproduction and Sources

Table 8: Current project scripts and outputs

Directory or entry point	Purpose
`trim/run_trim.m`	Trim residuals, 60 s unforced simulation and integration check
`A_B_Matrix/run_linear.m`	Central-difference A/B, subsystems, eigenvalues and local validation
`Gqde/run_Gqde.m`	Extraction of the Group 8 $G_{q\delta_e}$ and frequency-response verification
`AI_draft/loop.m`	Feedback transfer relationships for a supplied signed compensator
`AI_draft/report_english_version.tex`	Single English LaTeX source, including a native TikZ block diagram
`AI_draft/trim.png`	Figure exported from the actual trim simulation

Printed matrices and coefficients are rounded. Further numerical work should use the full-precision MAT/JSON files.

References

[1] AERO5560, *Group Project: Aircraft Closed-Loop Control System Design and Analysis*, assignment brief, pp. 1–5.

[2] *Group8 AircraftData*: `FlightDataAircraft8.p`, `ICsAircraft8_200knts_300ft.mat`, and `Note.txt`.

[3] AERO5560, *Flight Simulation Student Version*, supplied nonlinear functions and interface documentation.

[4] AERO5560, Lectures 1-2, 2-1/2, 7-2, 8-1/2, 9-1/2 and 10-1: linearisation, transfer functions, feedback architectures and controller design.

A Full Twelve-State Matrices

State and input ordering follows Section 1. The matrices below retain approximately eight significant figures. Each element has units given by its row's state-derivative units divided by its column-variable units.

$$A = \begin{bmatrix}
-0.061443341 & 0 & 0.05552868 & 0 & -0.47243227 & 0 & 0 & -9.8098966 & 0 & 0 & 0 & 3.1754685 \times 10^{-5} \\
0 & -0.37729298 & 0 & 0.53108604 & 0 & -102.86881 & 9.8487485 & 0 & 0 & 0 & 0 & 0 \\
-0.16960655 & 0 & -4.7873124 & 0 & 99.448231 & 0 & 0 & -0.045044714 & 0 & 0 & 0 & -0.00094886847 \\
0 & -0.20268929 & 0 & -13.808594 & 0 & 1.4524332 & 0.019116101 & 0 & 0 & 0 & 0 & 0 \\
0.00024757942 & 0 & -0.11258862 & 0 & -3.6747618 & 0 & 0 & 0 & 0 & 0 & 0 & -1.3364074 \times 10^{-6} \\
0 & 0.058710969 & 0 & -0.54309133 & 0 & -1.0812624 & -0.0033528581 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0.0045917624 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1.0000105 & 0 & 0 & 0 & 0 & 0 & 0 \\
0.99998946 & 0 & 0.004591714 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & -0.47243227 & 0 & 102.888 & 0 & 0 & 0 \\
-0.004591714 & 0 & 0.99998946 & 0 & 0 & 0 & 0 & -102.888 & 0 & 0 & 0 & 0
\end{bmatrix}$$

A.1 Full Input Matrix

$$B = \begin{bmatrix} 2.7631804 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3.7014393 & 20.777334 & 0 \\ -5.9699623 \times 10^{-5} & -62.906581 & 0 & 0 & 0 \\ 0 & 0 & -318.33488 & -3.5101752 & 0 \\ -1.7950804 \times 10^{-5} & -88.196939 & 0 & 0 & 0 \\ 0 & 0 & -16.168938 & -27.867108 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{39}$$

B Additional Subsystem Matrices

B.1 Five-State Longitudinal Model with Altitude

States: $[u, w, q, \theta, Z_e]$; inputs: $[\delta_T, \delta_e]$.

$$A_{\ell,5} = \begin{bmatrix} -0.061443341 & 0.05552868 & -0.47243227 & -9.8098966 & 3.1754685 \times 10^{-5} \\ -0.16960655 & -4.7873124 & 99.448231 & -0.045044714 & -0.00094886847 \\ 0.00024757942 & -0.11258862 & -3.6747618 & 0 & -1.3364074 \times 10^{-6} \\ 0 & 0 & 1 & 0 & 0 \\ -0.004591714 & 0.99998946 & 0 & -102.888 & 0 \end{bmatrix} \tag{40}$$

$$B_{\ell,5} = \begin{bmatrix} 2.7631804 & 0 \\ -5.9699623 \times 10^{-5} & -62.906581 \\ -1.7950804 \times 10^{-5} & -88.196939 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \tag{41}$$

B.2 Five-State Lateral-Directional Model

States: $[v, p, r, \phi, \psi]$; inputs: $[\delta_a, \delta_r]$.

$$A_{\text{lat}} = \begin{bmatrix} -0.37729298 & 0.53108604 & -102.86881 & 9.8487485 & 0 \\ -0.20268929 & -13.808594 & 1.4524332 & 0.019116101 & 0 \\ 0.058710969 & -0.54309133 & -1.0812624 & -0.0033528581 & 0 \\ 0 & 1 & 0.0045917624 & 0 & 0 \\ 0 & 0 & 1.0000105 & 0 & 0 \end{bmatrix} \tag{42}$$

$$B_{\text{lat}} = \begin{bmatrix} 3.7014393 & 20.777334 \\ -318.33488 & -3.5101752 \\ -16.168938 & -27.867108 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \tag{43}$$

C Complete Extracted Aerodynamic Parameters

Original field names and nondimensional aerodynamic conventions are retained; dimensional A/B matrices are obtained by numerical linearisation.

Table 9: Original Group 8 aerodynamic fields

Field	Value	Field	Value
alpha_o	−0.013772917	Cmad	0.9531
Cdo	0.020342	Cmde	−2.3941614
k	0.063776	Cnb	0.027705
CLa	4.786	Cnbd	−0.0051
CLq	9.269	Cnp	−0.000421
CLad	−1.1439	Cnr	−0.140618
CLde	0.61398157	Cnda	−0.015126086
CLo	0.065917179	Cndr	−0.11229973
Cyb	−0.379162	Clb	−0.024446
Cybd	0.117	Clbd	0.0068
Cyp	0.016304	Clp	−0.456441
Cyr	0.1222	Clr	0.0601
Cyda	0.036153637	Clda	−0.35735378
Cydr	0.20294165	Cldr	0.000394
Cmo	0.03337	CDdf	0
Cma	−0.295926	CLdf	0
Cmq	−25.094866	Cmdf	0